

■ Crop Recommendation System

Using Machine Learning Classification

Course:	Fundamentals of AI and ML
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1. Abstract

Agriculture is the backbone of the Indian economy, yet millions of farmers continue to make crop planting decisions based solely on tradition and intuition. This project presents a machine learning-based Crop Recommendation System that helps farmers decide which crop to plant by analyzing key soil and environmental parameters. Using a Random Forest Classifier trained on the Kaggle Crop Recommendation Dataset (2200 samples, 22 crops), the system achieves approximately 99% accuracy. Given inputs such as nitrogen, phosphorus, potassium levels, temperature, humidity, soil pH, and rainfall, the model predicts the most suitable crop and returns a confidence score.

2. Problem Statement

India is one of the world's largest agricultural nations, yet crop failure and low yield remain persistent issues. A primary reason is that farmers — especially in rural areas — do not have access to expert agronomic guidance and rely on guesswork when selecting crops for a particular season or soil type.

The consequences of a wrong crop selection include:

- Financial losses due to poor harvest
- Soil degradation from growing unsuitable crops
- Water wastage from crops needing different irrigation than available
- Loss of livelihood for smallholder farmers

This project addresses the question: Given measurable soil and climate data, can a machine learning model accurately recommend the best crop to plant?

3. Why This Problem Matters

According to the Indian government, agriculture employs over 54% of the workforce and contributes significantly to the national GDP. States like Madhya Pradesh are heavily agricultural, and uninformed crop selection is a documented challenge in rural districts. A simple, data-driven recommendation tool can:

- Help individual farmers maximize yield per acre
- Reduce dependency on expensive agronomists
- Be deployed as a mobile-friendly app for remote access
- Scale to support government agricultural advisory systems

4. Dataset

Attribute	Details
Source	Kaggle – Crop Recommendation Dataset

Rows	2200 samples
Crop Classes	22 (Rice, Wheat, Maize, Cotton, etc.)
Features	N, P, K, Temperature, Humidity, pH, Rainfall
Target	Crop Label (string)
Missing Values	None

The dataset is clean, balanced, and well-labeled — making it ideal for a classification task. Each of the 22 crop classes has exactly 100 samples, ensuring no class imbalance issues.

5. Approach and Methodology

5.1 Exploratory Data Analysis (EDA)

Before building the model, the dataset was explored visually to understand feature distributions and correlations:

- Plotted histograms of each input feature grouped by crop type
- Generated a correlation heatmap to identify redundant or related features
- Created a bar chart showing distribution of all 22 crop classes
- Observed that rainfall and humidity were highly separating features

5.2 Feature Engineering

No additional feature engineering was needed due to the clean nature of the dataset. All 7 input features (N, P, K, temperature, humidity, pH, rainfall) were used directly. The target column (label) was label-encoded for model training.

5.3 Model Selection

Several classification algorithms were evaluated:

Algorithm	Accuracy	Notes
Decision Tree	~90%	Prone to overfitting
Random Forest	~99%	Best performance — selected
K-Nearest Neighbors	~97%	Slower at prediction time
Naive Bayes	~83%	Assumes feature independence

5.4 Training and Evaluation

The dataset was split into 80% training and 20% testing. The Random Forest Classifier was trained using Scikit-learn with default hyperparameters and achieved ~99% accuracy on the test set. Evaluation metrics used included accuracy score, confusion matrix, and classification report (precision, recall, F1-score per class).

6. Core Implementation

The central prediction function used in this project:

```
from sklearn.ensemble import RandomForestClassifier
import pandas as pd

def recommend_crop(N, P, K, temperature, humidity, ph, rainfall):
    input_data = [[N, P, K, temperature, humidity, ph, rainfall]]
    prediction = model.predict(input_data)
    probability = model.predict_proba(input_data).max() * 100
    return prediction[0], probability

# Example usage (from submitted code):
my_crop, confidence = recommend_crop(25, 55, 65, 18, 45, 7.5, 80)
print(f"Your crop: {my_crop} ({confidence:.1f}%)" )
```

The function accepts the 7 soil/climate parameters, passes them through the trained Random Forest model, and returns both the predicted crop name and the model's confidence percentage.

7. Results and Visualizations

The following visualizations were generated as part of the analysis:

- Confusion Matrix — showing near-perfect classification across all 22 crops
- Feature Importance Plot — Rainfall, humidity, and K were the top 3 features
- Crop Distribution Bar Chart — confirmed balanced class distribution
- Correlation Heatmap — N, P, K showed mild positive correlation

Metric	Value
Model	Random Forest Classifier
Test Accuracy	~99%
Training Set Size	1760 samples (80%)
Test Set Size	440 samples (20%)
Number of Classes	22 crops
Top Feature	Rainfall

8. Challenges Faced

8.1 Model Selection

Initially, a Decision Tree was used, which showed signs of overfitting on training data. Switching to Random Forest — an ensemble of decision trees — resolved this and generalized much better to unseen test samples.

8.2 Understanding Feature Importance

Determining which features matter most required plotting and analyzing feature importance scores from the model. It was initially surprising that rainfall outweighed soil nutrients in importance, but this was confirmed by domain knowledge about crop-water dependency.

8.3 Making the Output Useful

Simply outputting a crop name was not enough. Adding a confidence score using `predict_proba()` made the output actionable — a farmer can see not just the recommendation but also how confident the model is.

9. Key Learnings

This project reinforced and deepened several concepts from the Fundamentals of AI and ML course:

- **Classification vs Regression:** This is a multiclass classification problem (22 output classes), not regression — understanding this distinction was fundamental.
- **Ensemble Methods:** Random Forest uses bagging to reduce variance. This project gave hands-on experience with why ensembles outperform single models.
- **Model Evaluation:** Using accuracy alone is insufficient. Precision, recall, and the confusion matrix give a more complete picture of model performance.
- **EDA is critical:** Time spent on exploratory data analysis directly informed better model choices and helped identify the most important features.
- **Real-world framing matters:** Grounding the ML solution in a real problem (Indian agriculture) made the project more meaningful and the report easier to write.

10. Future Scope

While the current system works well as a proof of concept, several improvements are possible:

- Deploy as a web app using Flask or Streamlit for real farmer usage
- Integrate live weather API data to auto-fill temperature and rainfall
- Add a fertilizer recommendation module as a companion feature
- Collect real soil test data from local farmers to improve generalization
- Build a mobile-friendly interface in regional languages (Hindi, Marathi)

11. Conclusion

The Crop Recommendation System demonstrates how a relatively simple machine learning classification model can solve a real, impactful problem. With ~99% accuracy on the test dataset and a clean, interpretable output (crop name + confidence score), the system is ready to serve as the foundation of a practical agricultural advisory tool.

This project successfully applied core concepts from the Fundamentals of AI and ML course — including supervised learning, classification, model evaluation, and data visualization — to a

problem observed in the real world. It reflects not just technical implementation, but thoughtful problem identification and purposeful solution design.

12. References

- Kaggle Crop Recommendation Dataset — <https://www.kaggle.com/datasets/atharvaingle/crop-recommendation-dataset>
- Scikit-learn Documentation — <https://scikit-learn.org/stable/>
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- Breiman, L. (2001). Random Forests. Machine Learning, 45(1), 5-32.
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